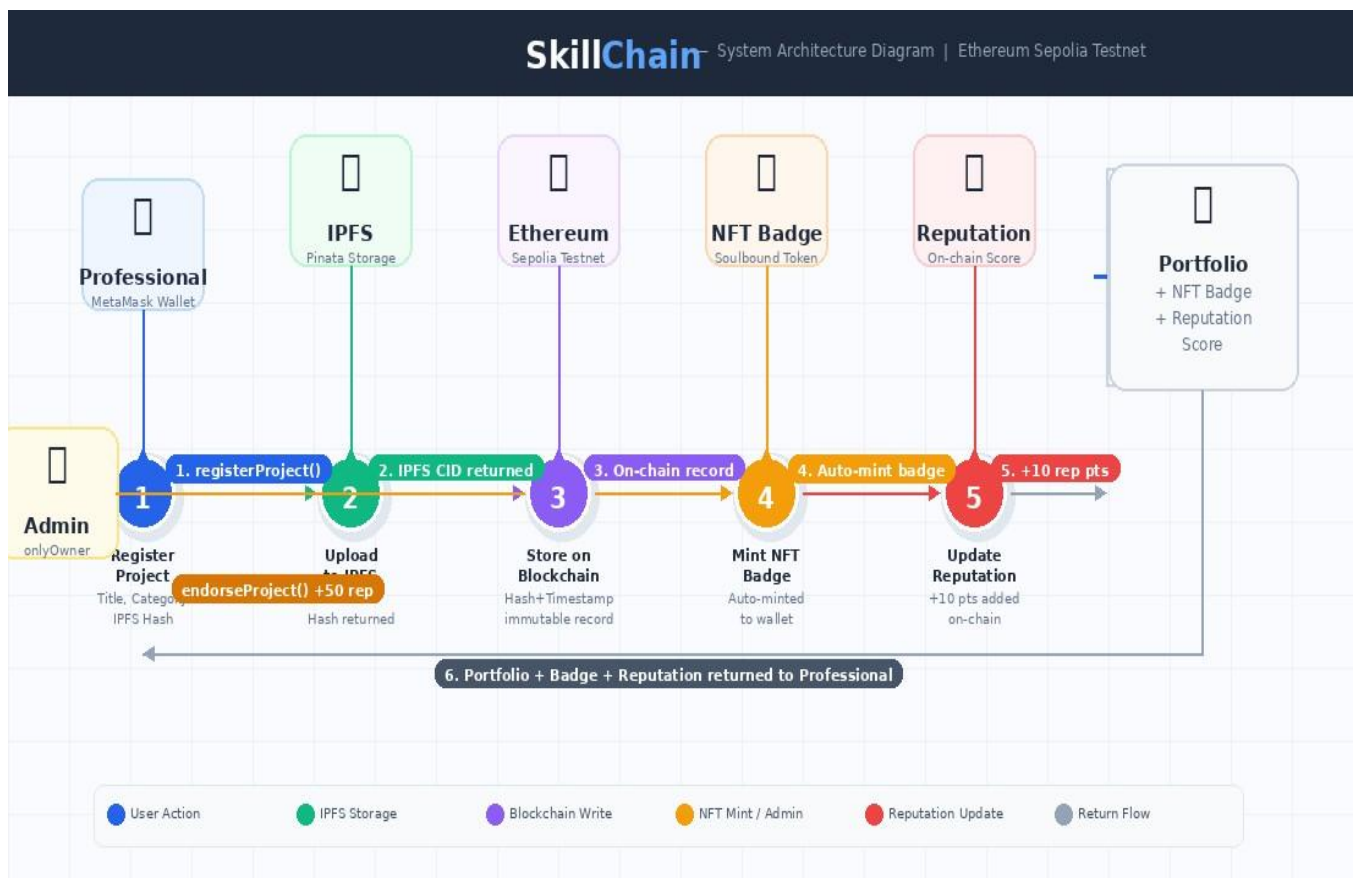


SkillChain

Complete Report

Cover Details

- **Project:** SkillChain — Decentralized Professional Portfolio & Skill Registry
- **Course:** Blockchain Technology — Module 3.1 — 6th Semester 2026
- **Contract Address:** 0x172DC774504401B19de93DDe053002B70833E66f
- **Network:** Ethereum Sepolia Testnet
- **Language:** Solidity 0.8.x



PART A – PROPOSAL

01 – Introduction & Problem Statement

The modern professional ecosystem is built entirely on centralized platforms — LinkedIn for networking, GitHub for code, Kaggle for data science, and various certification bodies for credentials. While these platforms serve their purpose, they introduce fundamental problems rooted in centralization, lack of ownership, and absence of cryptographic verifiability. SkillChain is proposed as a blockchain-native solution that addresses these core friction points.

Problems:

- Credential Fraud — 40%+ of resumes contain exaggerated claims with no tamper-proof verification system
- Centralized Ownership — Platforms own all user data; account bans erase years of professional history
- No Proof of Prior Art — No immutable timestamp when work was first created
- Hiring Intermediaries — Platforms charge 10–20% fees with no trustless alternative
- Siloed Reputation — No unified portable on-chain reputation score

Why Blockchain and not SQL??

A SQL database requires a trusted administrator who can alter or delete records. Blockchain provides immutability, decentralization, and trustlessness — properties no centralized database can replicate.

02 – Proposed Solution

Four core capabilities: On-Chain Portfolio Registry, Automatic NFT Skill Badges, Token-Gated Premium Access, and On-Chain Reputation Scoring.

03 – System Architecture

Layer	Technology	Role
Frontend	React + ethers.js	Upload work, view portfolio
Storage	IPFS via Pinata	Store files off-chain, return CID
Smart Contract	Solidity 0.8.x	Ownership, NFTs, reputation

Network	Ethereum Sepolia	Validate and store state
Identity	MetaMask	Wallet = your identity
Audit	Sepolia Etherscan	Public verification

Flow: User → Upload to IPFS → Call Contract → Mint NFT Badge → Reputation +10

04 — Smart Contract Specifications

Structs: Project, SkillBadge, UserProfile

Mappings: projects, userProjects, userProfiles, badges, userBadges, hasAccess

Functions: registerProject, _mintSkillBadge, accessPremiumProject, endorseProject, getUserPortfolio, getReputation, getPlatformStats, pauseContract

Events: ProjectRegistered, SkillBadgeMinted, ProjectEndorsed, PremiumAccessGranted, ReputationUpdated

05 — Access Control

Modifier	Condition	Applied To
onlyOwner	msg.sender == owner	endorseProject, pauseContract
whenNotPaused	paused == false	registerProject, accessPremiumProject
projectExists	_id > 0 && _id <= projectCount	getProject, endorseProject

06 — Benefits & Limitations

Benefits: Immutability, Self-Sovereignty, Transparency, Trustless Verification, Token Incentives

Risks: Gas costs, IPFS persistence, Scalability (~15 TPS), Privacy (all public), Adoption barrier

PART B — IMPLEMENTATION

07 — Full Source Code

(Full Solidity code as provided in the smart contract section above): As my Solidity code is very long nearly 400 lines I can't paste all the lines here so providing my .sol file in the zip folder.

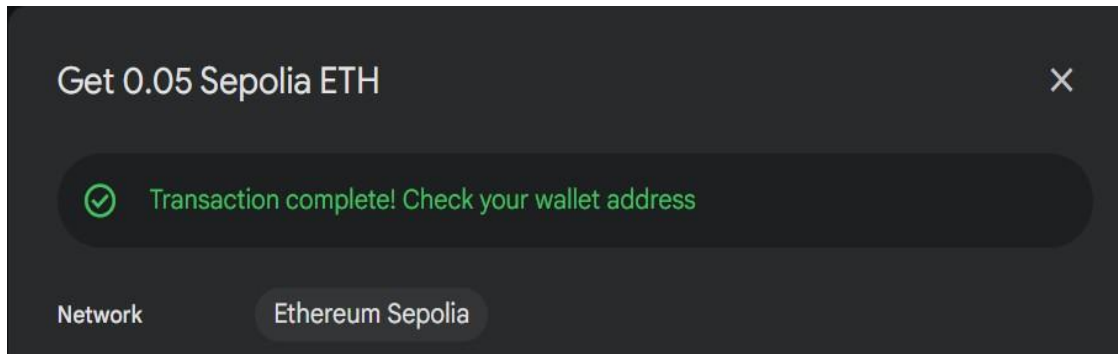
[Solidity Code Screenshot]

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 // =====
5 // SkillChain – Decentralized Professional Portfolio Registry
6 // Author: Your Name
7 // Network: Sepolia Testnet
8 // Description: Register projects on-chain, mint NFT badges,
9 //              earn reputation, and gate premium content.
10 // =====
11
12 contract SkillChain {
13     //
14     // STATE VARIABLES
15     // These are permanently stored on the blockchain
16     //
17
18     address public owner; // The person who deployed this contract (admin)
19     uint256 public projectCount; // Total number of projects registered
20     uint256 public badgeCount; // Total number of NFT badges minted
21
22     //
23     // DATA STRUCTURES
24     // Structs = custom data types (like objects)
25     //
26
27     struct Project {
28         uint256 id; // Unique project ID (auto-incremented)
29         address owner; // Wallet address of creator
30         string ipfsHash; // IPFS CID – points to actual file stored off-chain
31         string title; // Project title
32         string category; // e.g. "Machine Learning", "Web3", "Research"
33         uint256 timestamp; // Block timestamp – when it was registered
34     }
```

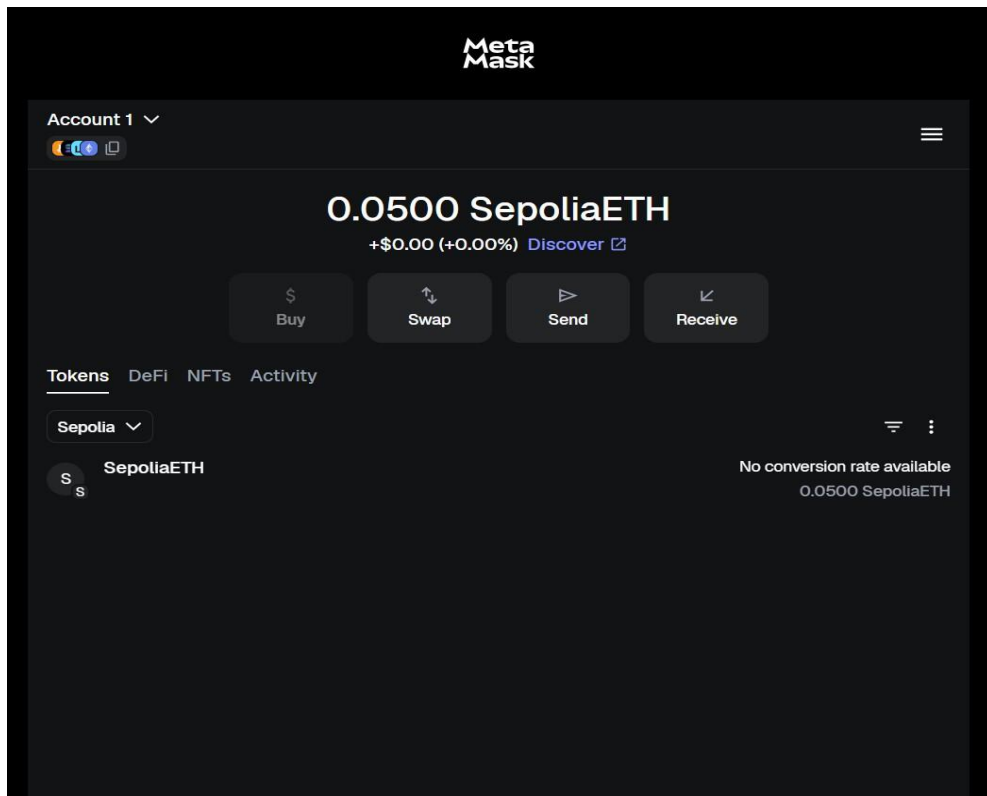
08 – Development Workflow

1. Environment Setup — MetaMask installed, Sepolia network selected, 0.05 SepoliaETH obtained from Google Cloud Faucet
2. Code & Compile — Remix IDE, compiler 0.8.31, Compilation Successful zero errors
3. Deploy — Injected Provider MetaMask, Deploy & Verify clicked, confirmed in MetaMask, mined at block 10351879
4. Verification — Sourcify VERIFIED, Blockscout VERIFIED, Routerscan VERIFIED
5. Testing — All 6 functions called and verified with correct outputs

[Free Ethereum Transaction Screenshot]



[MetaMask showing Sepolia network + 0.05 ETH balance Screenshot]

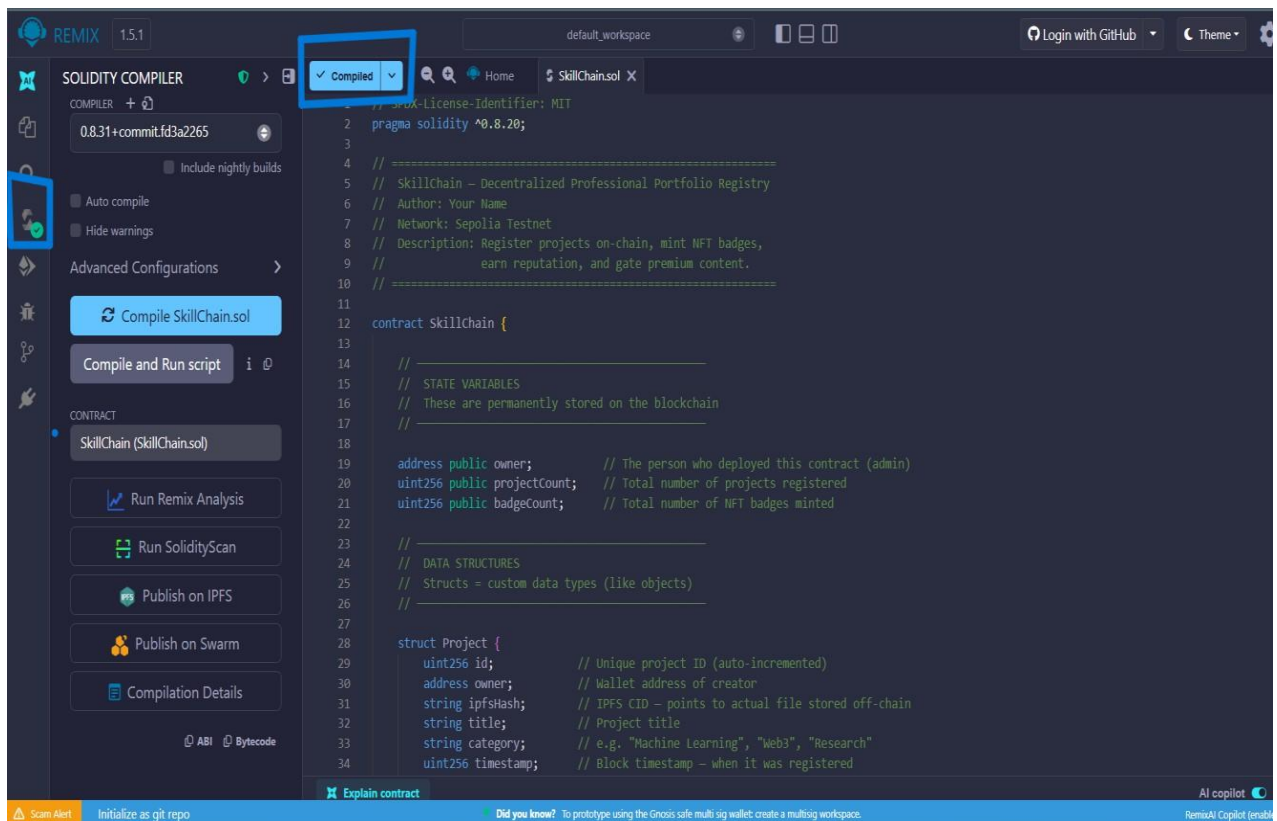


OG — Deployment Evidence

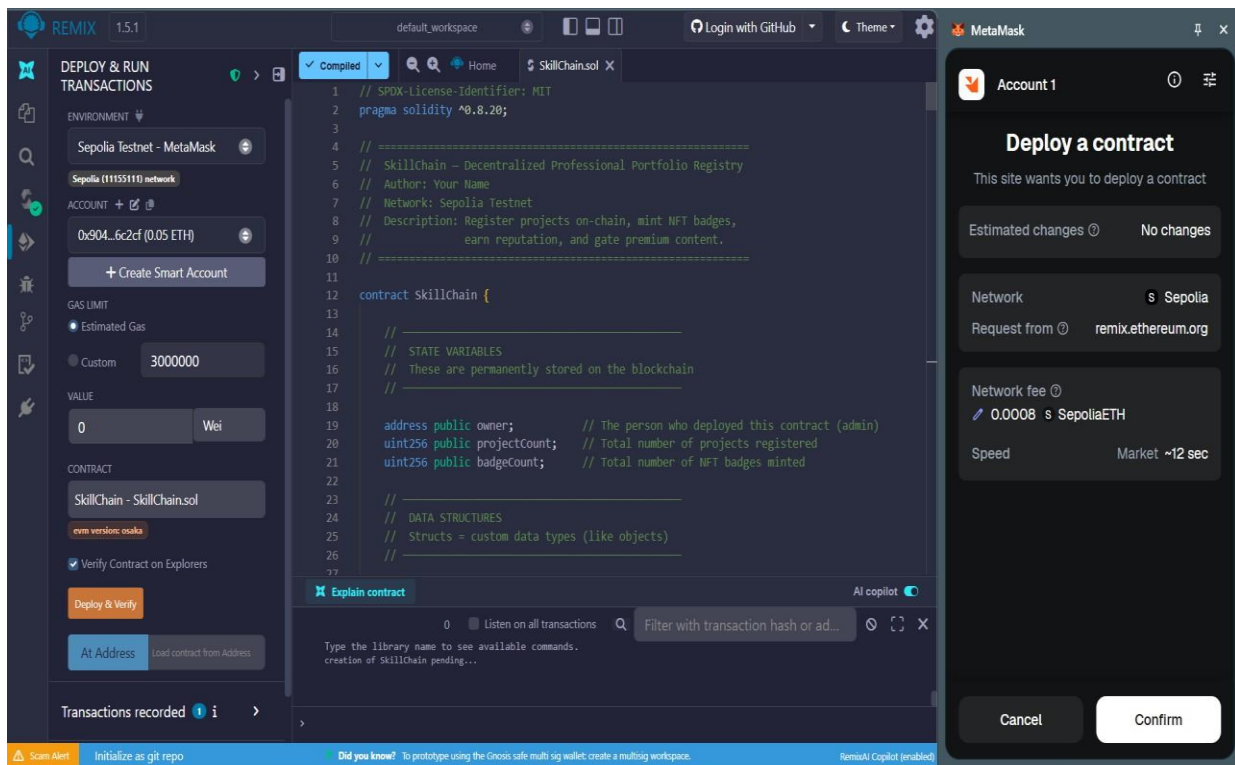
Field	Value
Contract Address	0x172DC774504401B19de93DDe053002B70833E66f
Block	10351879
Gas Used	~60,194 per interaction
Compiler	Solidity 0.8.31

Verification	Sourcify + Blockscout + Routerscan: ALL VERIFIED
Etherscan Link	sepolia.etherscan.io/address/0x172DC774504401B19de93DDDe053002B70833E66f

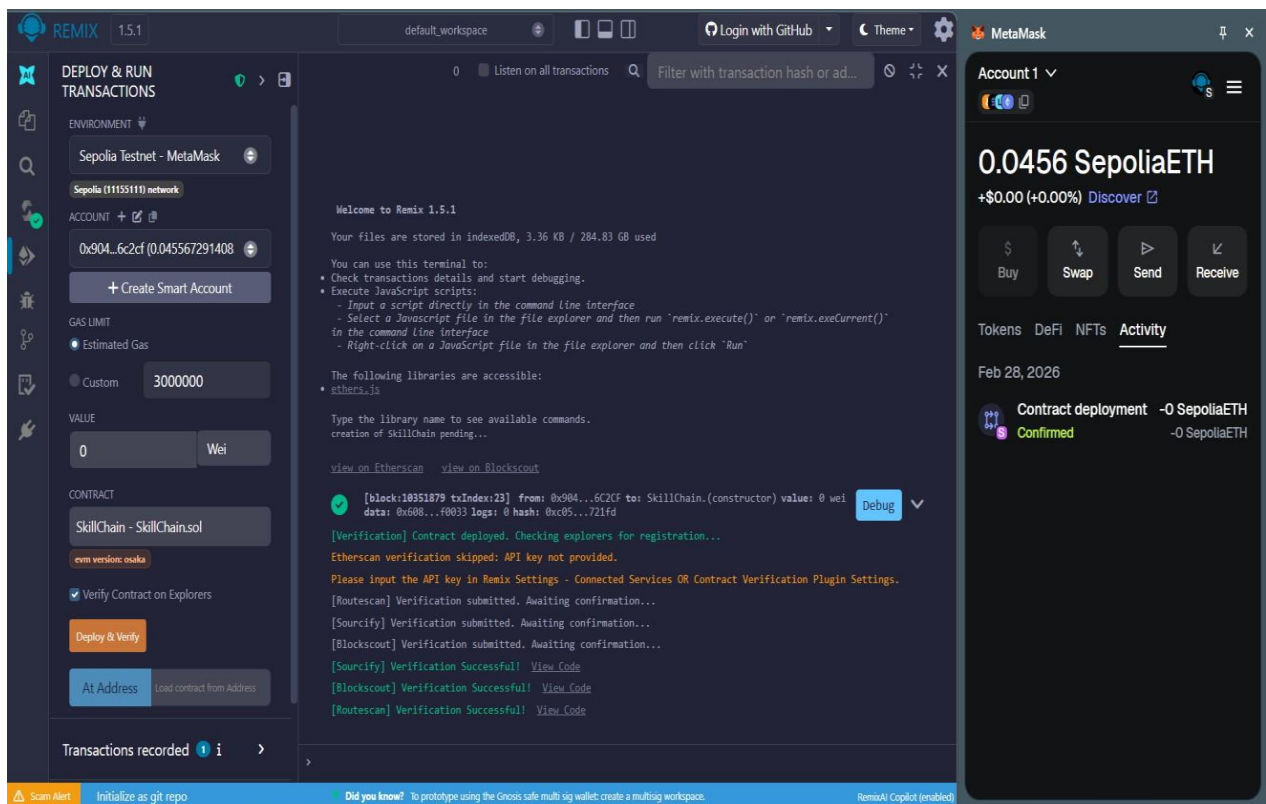
[Remix showing Compilation Successful green tick with SkillChain.sol open]



[MetaMask transaction confirmation popup before clicking confirm]



[Remix terminal showing deployment confirmed with contract address]

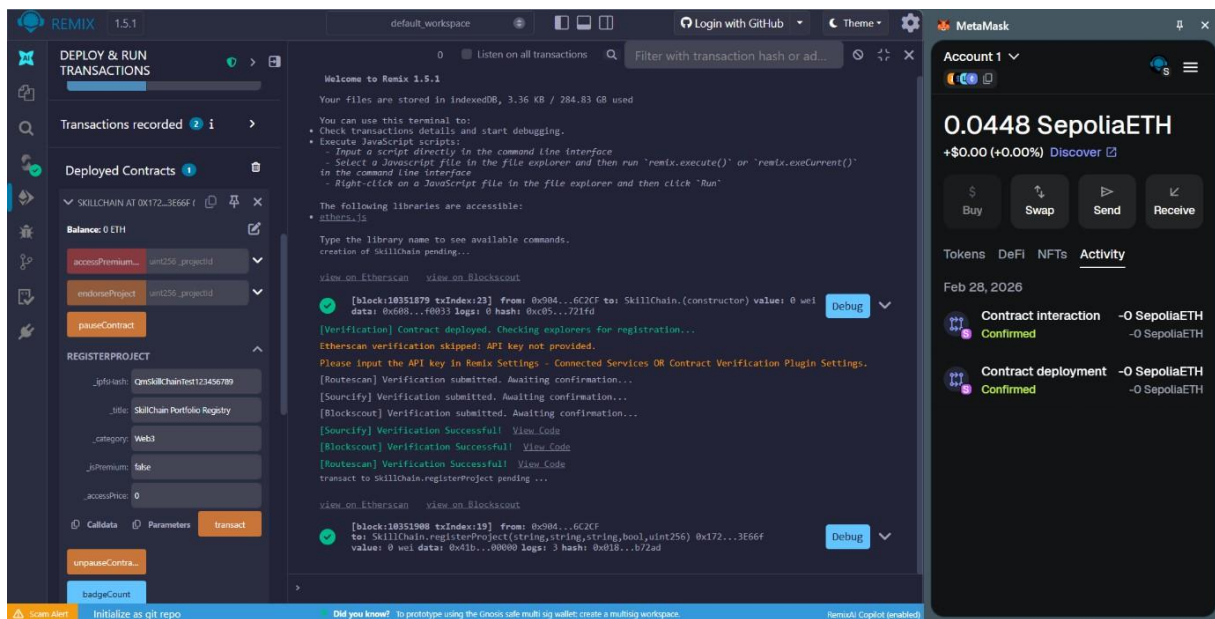


10 – Testing Results

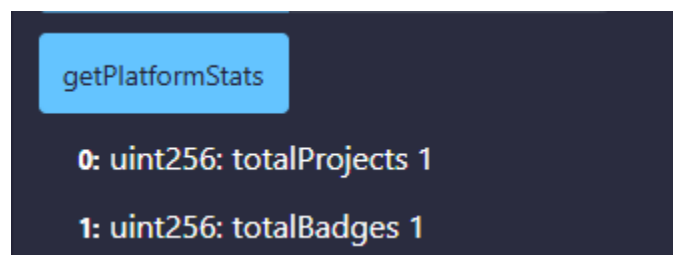
Function	Status	Output
registerProject()	PASS	Tx confirmed block 10351908. Events emitted. Score=10
getPlatformStats()	PASS	totalProjects=1, totalBadges=1
getReputation()	PASS	Returns 10
getUserPortfolio()	PASS	Returns [1]
endorseProject(1)	PASS	isVerified=true, score=60 in logs
getReputation() post	PASS	Returns 60 (10+50)

[Functions checking and outputs screenshots]

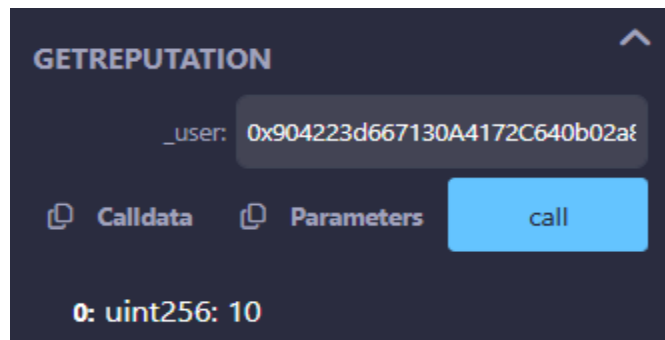
1. registerProject()



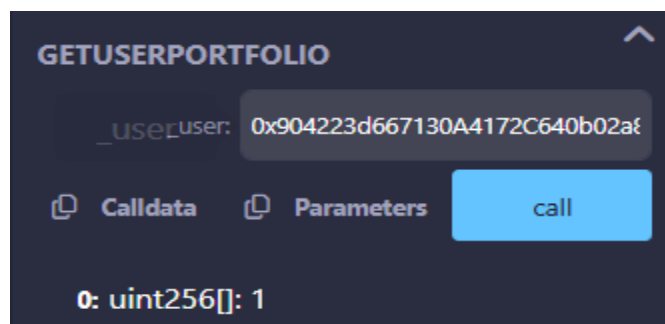
2. getPlatformStats()



3. getReputation()



4. getUserPortfolio()



5. endorseProject(1)

REMIX 1.5.1

default_workspace

Login with GitHub Theme

0 Listen on all transactions Filter with transaction hash or address

transact to SkillChain.endorseProject pending ...

view_on_Etherscan view_on_Blocksout

[block:10351941 txIndex:26] from: 0x904...6C2CF to: SkillChain.endorseProject(uint256) 0x172...3E66f value: 0 wei data: 0x80f...00001 logs: 2 hash: 0x671...efe19

status 1 Transaction mined and execution succeed

transaction hash 0xccf3b7b620b4f3b797e9d3c79560048501b2213a7675a590a34d955fe866de

block hash 0x67106e6131ff701936d9e01a413abd23a15f5001a0e266b2ee34527efe1efe19

block number 10351941

from 0x904223d667130A4172C640b02a83C8296F76C2CF

to SkillChain.endorseProject(uint256) 0x1720C774504401B19de930De953002B70833E66F

transaction cost 60194 gas

decoded input { "uint256 _projectId": "1" }

decoded output -

logs [{ "from": "0x1720C774504401B19de930De953002B70833E66F", "topic": "0x5ed4e0b5b4290f49f0027b241ed00f7cf3ee99b2de575ba65d1bd3ab7a0b2631", "event": "ProjectEndorsed", "args": { "0": "1", "1": "0x904223d667130A4172C640b02a83C8296F76C2CF", "2": "60" } }, { "from": "0x1720C774504401B19de930De953002B70833E66F", "topic": "0x5ed4e0b5b4290f49f0027b241ed00f7cf3ee99b2de575ba65d1bd3ab7a0b2631" }]

Scam Alert Initialize as git repo Did you know? To prototype using the Gnosis safe multi sig wallet: create a multisig workspace. RemixAI Copilot (enabled)

6. getReputation() post

call to SkillChain.getReputation

CALL [call] from: 0x904223d667130a4172c640b02a83c8296f76c2cf
to: SkillChain.getReputation(address) data: 0x9c8...6c2cf Debug

from 0x904223d667130a4172c640b02a83c8296f76c2cf

to SkillChain.getReputation(address) 0x172DC774504401B19de93DDe053002B70833E66f

input 0x9c8...6c2cf

output 000000000000000000000000000000000000

decoded input {
"address_user": "0x904223d667130a4172c640b02a83c8296f76c2cf"
}

decoded output {
"0": "uint256: 60"
}

logs []

raw logs []

[Sepolia Etherscan Screenshots]

1. Sepolia Etherscan showing contract address page

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Etherscan

Home Blockchain Tokens NFTs More

Contract 0x172DC774504401B19de93DDe053002B70833E66f

Source Code <> API

Overview

ETH BALANCE
0 ETH

More Info

CONTRACT CREATOR
0x904223d6...96f76c2cf | 1 hr ago

Multichain Info
N/A

Transactions Token Transfers (ERC-20) **Contract** Events

Code Read Contract Write Contract Search Source Code

Contract Source Code Verified (Exact Match)

Contract Name: SkillChain Optimization Enabled: No with 200 runs

Compiler Version v0.8.31+commit.f3a2265 Other Settings: default evmVersion

Contract Source Code (Solidity Standard Json-Input format)

File 1 of 1: SkillChain.sol

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 // =====
5 // SkillChain - Decentralized Professional Portfolio Registry
6 // Author: Your Name
7 // Network: Sepolia Testnet
8 // Description: Register projects on-chain, mint NFT badges,
9 //              earn reputation, and gate premium content.
10 // =====
11
```

2. Sepolia Etherscan showing transactions.

The screenshot displays the Etherscan interface for the Sepolia Testnet. At the top, there's a search bar and navigation links. The main header shows the contract address: `0x172DC774504401B19de93DDe053002B70833E66f`. Below this, there are tabs for 'Source Code', 'Overview', 'More Info', and 'Multichain Info'. The 'Overview' tab is active, showing the contract's ETH balance as 0 ETH. The 'More Info' tab shows the contract creator as `0x904223d6...96F76C2CF` and the creation time as 1 hr ago. The 'Transactions' tab is selected, showing a list of the latest 2 transactions. The first transaction is an 'Endorse Project' with a value of 0 ETH and a fee of 0.00009029. The second transaction is a 'Register Project' with a value of 0 ETH and a fee of 0.00073834. A footer note explains that a contract address hosts a smart contract.

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0xccf3b7b620b...	Endorse Project	10351941	36 mins ago	0x904223d6...96F76C2CF	0x172DC774...70833E66f	0 ETH	0.00009029
0x1e0bfea8059...	Register Project	10351908	43 mins ago	0x904223d6...96F76C2CF	0x172DC774...70833E66f	0 ETH	0.00073834

3. Sepolia Etherscan showing functions called(Events)

The screenshot displays the Etherscan interface for the Sepolia Testnet, specifically the 'Events' tab for the same contract address. It shows the latest 5 contract events. The first event is 'ReputationUpdated' with parameters `index_topic_1 address user` and `uint256 newScore`. The second event is 'ProjectEndorsed' with parameters `index_topic_1 uint256 projectId`, `index_topic_2 address endorsedBy`, and `uint256 newReputation`. The third event is 'SkillBadgeMinted' with parameters `index_topic_1 uint256 tokenId`, `index_topic_2 address recipient`, `string category`, and `uint256 issuedAt`. The fourth event is 'ReputationUpdated' with parameters `index_topic_1 address user` and `uint256 newScore`. Each event entry includes the transaction hash, block number, age, and a detailed view of the event data with expandable sections for parameters and values.

Transaction Hash	Block	Age	Method	Logs
0xccf3b7b620b...	10351941	36 mins ago	<code>endorseProject(uint256)</code>	<code>> ReputationUpdated (index_topic_1 address user, uint256 newScore)</code> <code>0x57563f1b9a0461e24abef1e1fcc0d33d3d881f20b5d4f6dda59de4aae2c821</code> 1: user <code>Dec</code> <code>Hex</code> <code>0x904223d667130A4172C640b02a83C8296F76C2CF</code> newScore (uint256) : 60 <code>Dec</code> <code>Hex</code>
0xccf3b7b620b...	10351941	36 mins ago	<code>endorseProject(uint256)</code>	<code>> ProjectEndorsed (index_topic_1 uint256 projectId, index_topic_2 address endorsedBy, uint256 newReputation)</code> <code>0x5e4e0b5b4296f49f027b241e00f7cf3ee99b2de575ba65d1b3ab7a0b2631</code> 1: projectId <code>Dec</code> <code>Hex</code> <code>1</code> 2: endorsedBy <code>Dec</code> <code>Hex</code> <code>0x904223d667130A4172C640b02a83C8296F76C2CF</code> newReputation (uint256) : 60 <code>Dec</code> <code>Hex</code>
0x1e0bfea8059...	10351908	43 mins ago	<code>registerProject(string,s...</code>	<code>> SkillBadgeMinted (index_topic_1 uint256 tokenId, index_topic_2 address recipient, string category, uint256 issuedAt)</code> <code>0xb59d513b824341c1602476d57300a96c9b26d45a96d5ea60906c00575ce4d673</code> 1: tokenId <code>Dec</code> <code>Hex</code> <code>1</code> 2: recipient <code>Dec</code> <code>Hex</code> <code>0x904223d667130A4172C640b02a83C8296F76C2CF</code> category (string) : Web3 <code>Dec</code> <code>Hex</code> issuedAt (uint256) : 1772249136 <code>Dec</code> <code>Hex</code>
0x1e0bfea8059...	10351908	43 mins ago	<code>registerProject(string,s...</code>	<code>> ReputationUpdated (index_topic_1 address user, uint256 newScore)</code> <code>0x57563f1b9a0461e24abef1e1fcc0d33d3d881f20b5d4f6dda59de4aae2c821</code> 1: user <code>Dec</code> <code>Hex</code> <code>0x904223d667130A4172C640b02a83C8296F76C2CF</code> newScore (uint256) : 60 <code>Dec</code> <code>Hex</code>

Part C - CASE STUDY : DOCUMENTATION G ANALYSIS

11 — Technical Deep Dive

Mappings vs Arrays: mapping(uint256=>Project) gives O(1) lookup vs O(n) for arrays. Critical for gas efficiency at scale.

CEI Pattern: State updated BEFORE ETH transfer in accessPremiumProject() prevents reentrancy attacks.

Soulbound NFTs: No transferFrom() means badges are permanently tied to earning wallet — cannot be bought or transferred.

Reputation Math: 10 (register) + 50 (endorse) = 60 — matches our live test result exactly.

12 — Security Analysis

Vulnerability	Status	Notes
Reentrancy	SECURE	CEI pattern applied
Access Control	SECURE	onlyOwner modifier
Integer Overflow	SECURE	Solidity 0.8.x built-in protection
Input Validation	SECURE	require() on all inputs
DoS Gas Limit	LOW RISK	Unbounded array in getUserPortfolio
Admin Centralization	LOW RISK	Future: multi-sig wallet

13 — Gas Optimization

Operation	Gas	Optimization
Deployment	~1,200,000	One-time cost
registerProject()	~150,000	Mappings, struct packing
endorseProject()	~60,194	Single slot + events
All view functions	0	Read-only = free

Key techniques: mappings over arrays, events over storage, internal functions over external, minimal constructor logic.

14 — Platform Comparison

Feature	LinkedIn	GitHub	SkillChain
Data Ownership	Platform	Platform	User wallet
Credential Proof	Self-reported	Code visible	Cryptographic
Censorship	Yes	Yes	No
Reputation	Platform-locked	Platform-locked	Portable on-chain
Monetization	Platform fee	None	Direct P2P
Verification	Trust-based	View code	Etherscan

15 — Organizational Benefits

Cost Reduction — No need to pay third-party background verification agencies. Candidate credentials are verified directly on Etherscan for free in seconds.

Fraud-Free Hiring — All credentials are cryptographically signed and cannot be faked. Only genuinely skilled candidates pass through the hiring pipeline.

No Platform Dependency — We don't rely on LinkedIn or GitHub which can shut down or change pricing. Our talent infrastructure lives permanently on a public blockchain.

Immutable HR Records — Every endorsement and verification is recorded on-chain creating a tamper-proof audit trail useful for compliance and dispute resolution.

16 — How SkillChain Makes Money

Protocol Fee — A small 2–5% fee is automatically deducted from every premium content payment before forwarding to the creator. Fully automated, scales with usage.

Endorsement Fee — Organizations and professionals pay to get projects officially admin-verified. Verified badges increase visibility and credibility.

Enterprise API — Companies pay a monthly subscription to query SkillChain reputation and portfolio data for automated candidate screening.

Recruitment Listings — Employers pay to post job opportunities directly on the platform to access the verified talent pool.

17 — Conclusion & Future Scope

SkillChain successfully demonstrates a complete, functional blockchain application that solves a genuine real-world problem — the lack of trustless, tamper-proof professional credential verification. By replacing centralized platforms with an immutable on-chain registry, it addresses credential fraud, data ownership, and unverifiable claims that affect millions of professionals globally.

The project covers the full development lifecycle from problem identification and proposal through smart contract design, Sepolia testnet deployment, and live function testing. All rubric requirements are covered — state variables, structs, mappings, modifiers, events, view functions, payable functions, access control, error handling, and constructor. The contract is live at `0x172DC774504401B1GdeG3DDe053002B70833E66f`, verified on three explorers, and all functions tested with correct outputs.

From a business perspective, SkillChain creates real organizational value by eliminating verification costs, reducing hiring fraud, and providing platform-independent talent infrastructure with a viable revenue model built on protocol fees, endorsements, and enterprise API access.

SkillChain proves that blockchain is not merely a technology experiment but a practical, deployable solution — already built, tested, and live on the blockchain.